King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 208 Major Exam I Term 251

Original exam date: 30 September 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. At 4:00 pm, a thermometer reading 20°C is put into a freezer where the temperature is −10°C. If the reading is 5°C at 4:02 pm, then the reading at 4:06 pm is

(a) −6.25°C

(b) −4.25°C

(c) 0°C

(d) −7.25°C

(e) −8.25°C

2. If the function $k(x)$ with $k(0) = 0$ makes

$$\frac{dy}{dx} = \frac{y\cos x + 2xe^y + 3}{k(x) - x^2e^y + 2x}$$

an exact differential equation, then $k(x) =$

(a) $-\sin x - 2x$

(b) $-\sin x + 4x$

(c) $\sin x - 2x$

(d) $\sin x + 2x$

(e) $-\sin x + 3x$

3. If the solution space of the system

$$\begin{aligned} x_1 - 4x_2 - 3x_3 - 7x_4 &= 0 \\ 2x_1 - x_2 + x_3 + 7x_4 &= 0 \\ x_1 + 2x_2 + 3x_3 + 11x_4 &= 0 \end{aligned}$$

is the set of all linear combinations of the form $s\mathbf{u} + t\mathbf{v}$ where s, t are real numbers, then

(a) $\mathbf{u} = (-5, -3, 0, -1)$ and $\mathbf{v} = (-1, -1, 1, 0)$

(b) $\mathbf{u} = (5, -3, 0, 1)$ and $\mathbf{v} = (-1, -1, 1, 0)$

(c) $\mathbf{u} = (-5, -3, 0, 1)$ and $\mathbf{v} = (-1, -1, 1, 0)$

(d) $\mathbf{u} = (-5, -3, 0, 1)$ and $\mathbf{v} = (1, -1, 1, 0)$

(e) $\mathbf{u} = (-5, -3, 0, -1)$ and $\mathbf{v} = (1, 1, -1, 0)$

4. The general solution of the linear differential equation

$$(x^2+1)\frac{dy}{dx} + 3x^3 y = 6x\, e^{-\frac{3}{2}x^2}$$

is

(a) $y(x) = \left[-4 + c(x^2+1)^{\frac{3}{2}}\right] e^{-\frac{3}{2}x^2}$

(b) $y(x) = \left[1 + c(x^2+1)^{\frac{3}{2}}\right] e^{-\frac{3}{2}x^2}$

(c) $y(x) = \left[-2 + c(x^2+1)^{\frac{3}{2}}\right] e^{-\frac{3}{2}x^2}$

(d) $y(x) = \left[-3 + c(x^2+1)^{\frac{3}{2}}\right] e^{-\frac{3}{2}x^2}$

(e) $y(x) = \left[-6 + c(x^2+1)^{\frac{3}{2}}\right] e^{-\frac{3}{2}x^2}$

5. The general solution of the differential equation

$$\left(2xy + 3y^2\right) dx - \left(2xy + x^2\right) dy = 0$$

is given by

(a) $y^2 + 2xy = cx^3$

(b) $y^2 + xy = cx^4$

(c) $y^2 + xy = cx^3$

(d) $y^3 + xy = cx^2$

(e) $y^2 - xy = cx^2$

6. The solution of the initial-value problem

$$\left(1 + x\cos^2 y + x + \cos^2 y\right) dx - x \sin y\, dy = 0, \quad y(1) = 0$$

is

(a) $\ln|x| + 2x + \tan^{-1}(\cos y) = 1 + \dfrac{\pi}{4}$

(b) $\ln|x| + 4x + \tan^{-1}(\cos y) = 1 + \dfrac{\pi}{4}$

(c) $\ln|x| + 5x + \tan^{-1}(\cos y) = 1 + \dfrac{\pi}{2}$

(d) $\ln|x| + 3x + \tan^{-1}(\cos y) = 1 + \dfrac{\pi}{3}$

(e) $\ln|x| + x + \tan^{-1}(\cos y) = 1 + \dfrac{\pi}{4}$

7. The general solution of the differential equation $x^2y'' + 3xy' = 2$ is given by (Note: A and B are constants, and $x > 0$)

(a) $y(x) = \ln x + \dfrac{A}{x^3} + B$

(b) $y(x) = 2\ln x + \dfrac{A}{x^2} + B$

(c) $y(x) = \ln x + \dfrac{A}{x^2} + B$

(d) $y(x) = \ln x + \dfrac{A}{x^2} + Bx$

(e) $y(x) = 2\ln x + \dfrac{A}{x^3} + B$

8. A particle is moving in a straight line with acceleration $a(t) = t^2 \ln t$ and an initial velocity $v(1) = 0$. The velocity at any time $t > 1$ is given by

(a) $\dfrac{t^4}{4}\ln t - \dfrac{t^4}{8} + \dfrac{1}{8}$

(b) $\dfrac{t^3}{3}\ln t - \dfrac{t^3}{8} + \dfrac{1}{8}$

(c) $\dfrac{t^4}{4}\ln t - \dfrac{t^4}{9} + \dfrac{1}{9}$

(d) $\dfrac{t^2}{2}\ln t - \dfrac{t^2}{8} + \dfrac{1}{8}$

(e) $\dfrac{t^3}{3}\ln t - \dfrac{t^3}{9} + \dfrac{1}{9}$

9. A general solution of the exact differential equation

$$\left(y^2 + 3x^2 - 2xy^3\right)dx - \left(1 - 2xy + 3x^2y^2\right)dy = 0$$

is

(a) $y + xy^2 + x^2y^3 - x^3 = c$

(b) $y - 2xy^2 + x^2y^3 + x^3 = c$

(c) $y - xy^2 + x^2y^3 + x^3 = c$

(d) $y - xy^2 - x^2y^3 - x^3 = c$

(e) $y - xy^2 + x^2y^3 - x^3 = c$

10. Which one of the following subsets is not a subspace of $\mathbb{R}^3$?

(a) The set of all vectors (x_1, x_2, x_3) such that $x_3 = 0$

(b) The set of all vectors (x_1, x_2, x_3) such that $x_1 + x_2 = 0$

(c) The set of all vectors (x_1, x_2, x_3) such that $x_1 + x_2 + x_3 = 0$

(d) The set of all vectors (x_1, x_2, x_3) such that $x_3 = 1$

(e) The set of all vectors (x_1, x_2, x_3) such that $x_3 = x_1 + x_2$

11. If the vectors $\mathbf{u} = (5, -4, 3)$, $\mathbf{v} = (-2, 0, 3)$ and $\mathbf{w} = (a, -8, 1)$ are linearly dependent, then $3a =$

(a) -42

(b) 42

(c) 36

(d) 40

(e) -40

12. By making a suitable substitution, the differential equation $\dfrac{dy}{dx} = 1 + e^{y-x+5}$ can be transformed into a separable differential equation

(a) $e^{-v}\,dv = x\,dx$

(b) $e^{-v}\,dv = 2\,dx$

(c) $e^{v}\,dv = 2\,dx$

(d) $e^{v}\,dv = dx$

(e) $e^{-v}\,dv = dx$

13. Let $\mathbf{v}_1 = (5, 3, 4)$, $\mathbf{v}_2 = (3, 2, 5)$ and $\mathbf{w} = (1, 0, -7)$ be three vectors in $\mathbb{R}^3$. If $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2$, then $ab =$

(a) 10

(b) 6

(c) -6

(d) 0

(e) -8

14. By using a suitable substitution, we can transform the differential equation

$$\left(2y\cos^2 x - 2\sqrt{y}\right)dx + x\cos^2 x\,dy = 0$$

into the linear differential equation

(a) $v' + \dfrac{2}{x}v = \dfrac{1}{x}\cos^2 x$

(b) $v' + \dfrac{1}{x}v = \dfrac{1}{x}\sec^2 x$

(c) $v' + \dfrac{2}{x}v = \dfrac{1}{x}\sec^2 x$

(d) $v' - \dfrac{2}{x}v = \dfrac{1}{x}\sec^2 x$

(e) $v' - \dfrac{1}{x}v = \dfrac{1}{x}\sec^2 x$

15. Let $y = x^m$ be a solution of the differential equation $xy^{(4)} + 6y''' = 0$, then the sum of all values of m is equal to

(a) 0

(b) 2

(c) 1

(d) -1

(e) 3

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$T = -10 + 30e^{-kt}$, $e^{-2k} = \frac{1}{2}$: $T(6) = -10 + \frac{30}{8} = -6.25°\text{C}$.
2	**(a)**	$N = -k + x^2e^y - 2x$; $M_y = N_x \Rightarrow k' = -\cos x - 2$, so $k = -\sin x - 2x$.
3	**(c)**	RREF: $x_1 = -x_3 - 5x_4$, $x_2 = -x_3 - 3x_4$: $\mathbf{u} = (-5, -3, 0, 1)$, $\mathbf{v} = (-1, -1, 1, 0)$.
4	**(c)**	$\rho = e^{\frac{3}{2}x^2}(x^2+1)^{-\frac{3}{2}}$, $\rho y = -2(x^2+1)^{-\frac{3}{2}} + c$.
5	**(c)**	Homogeneous, $v = y/x$: $xv' = \frac{v^2+v}{2v+1}$, $v^2 + v = cx$, so $y^2 + xy = cx^3$.
6	**(e)**	$(1+x)(1+\cos^2 y)dx = x\sin y\,dy$: $\ln\lvert x\rvert + x + \tan^{-1}(\cos y) = 1 + \frac{\pi}{4}$.
7	**(c)**	$p = y'$: $(x^3p)' = 2x$, $p = \frac{1}{x} + \frac{C}{x^3}$, so $y = \ln x + \frac{A}{x^2} + B$.
8	**(e)**	By parts: $v = \frac{t^3}{3}\ln t - \frac{t^3}{9} + C$, $v(1) = 0 \Rightarrow C = \frac{1}{9}$.
9	**(e)**	$F = xy^2 + x^3 - x^2y^3 - y$ ($g' = -1$); times -1: $y - xy^2 + x^2y^3 - x^3 = c$.
10	**(d)**	$x_3 = 1$ does not contain $\mathbf{0}$; the others are homogeneous linear equations.
11	**(d)**	$\det = 160 - 12a = 0 \Rightarrow a = \frac{40}{3}$, so $3a = 40$.
12	**(e)**	$v = y - x + 5$: $v' = y' - 1 = e^v$, so $e^{-v}\,dv = dx$.
13	**(c)**	$5a + 3b = 1$, $3a + 2b = 0$: $a = 2$, $b = -3$ (third checks), so $ab = -6$.
14	**(b)**	Bernoulli $n = \frac{1}{2}$, $v = \sqrt{y}$: $v' + \frac{1}{x}v = \frac{1}{x}\sec^2 x$.
15	**(a)**	$m(m-1)(m-2)(m+3) = 0$: $m = 0, 1, 2, -3$, sum $= 0$.